How object segmentation and perceptual grouping emerge in noisy variational autoencoders

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INTRODUCTION

Humans effortlessly perceive and segment objects in **complex and ambiguous** environments.

Deep neural networks struggle to capture these capabilities[1-3].

Variational autoencoders[4] code for disentangled factors of the data distribution, where individual units code for individual image elements. Evidence of this coding has been found in the primate brain[5].

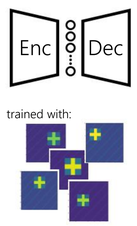
Here, we make variational autoencoders perform segmentation without training any additional parameters, and show that it results in **good segmentation quality** and **reproduces perceptual grouping**.



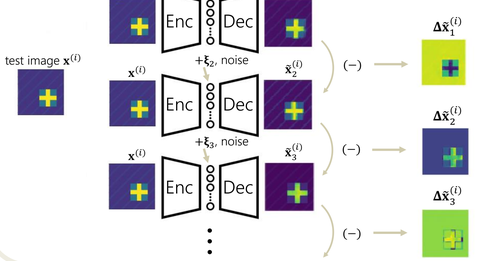
Image Credit: R. C. James

METHODOLOGY

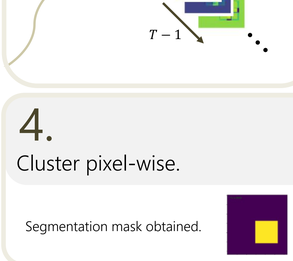
1. Train model to reconstruct images $\mathbf{x}^{(i)}$



2. Repeatedly inject i.i.d. noise to the latent layer, sampling outputs $\tilde{\mathbf{x}}_t^{(i)}$ from the model.
Record the differences in outputs $\Delta\tilde{\mathbf{x}}_t^{(i)}$ of two successive steps.



3. Stack the outputs $\Delta\tilde{\mathbf{x}}_t^{(i)}$.

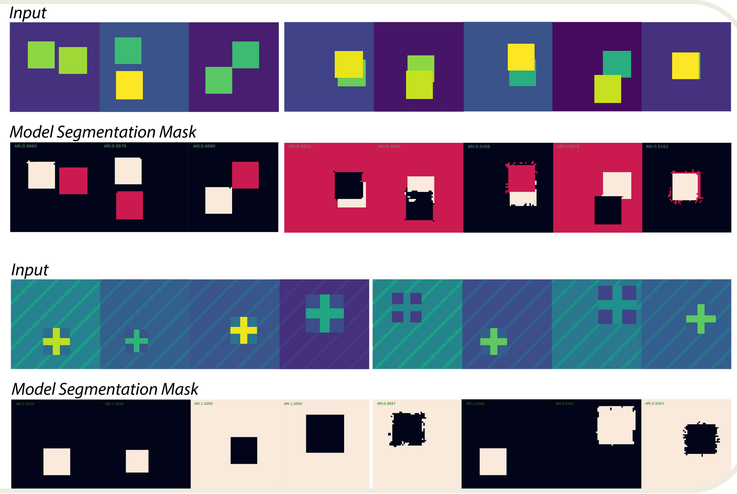


4. Cluster pixel-wise.

Segmentation mask obtained.

RESULTS: QUALITATIVE

Multi-object representation and occlusion



Perceptual grouping and illusory contours



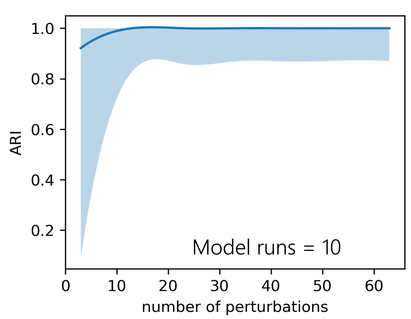
Generalization to complex stimuli: faces

1) Face segmentation
2) Binding of facial parts

RESULTS: QUANTITATIVE

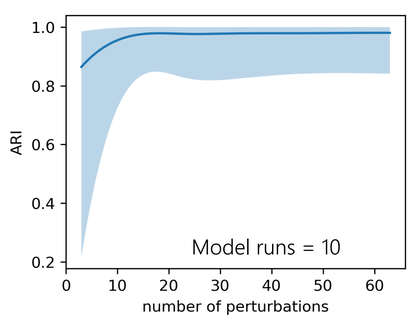
1. Illusory contours.





2. Occlusion.





CONCLUSION

It is possible to segment images without a dedicated learned computational module for segmentation.

Segmenting this way reproduces the human ability to handle **occlusions** and perceive **illusory contours**.

The method scales to more naturalistic images, and needs **only few recurrent iterations** to perform.

Humans could use a similar computation for automatically segmenting at relevant time-scales.

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References

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